

Student Management System

1. Introduction:

The Student Management System is a console-based application developed using C++. The main purpose of this system is to manage student records efficiently. It helps users to add, display, search, and delete student records easily using file handling operations.

This system reduces manual work and improves student record management by maintaining details such as Student ID, Student Name, and Marks in an organized manner.

2. Objectives:

- To maintain student records efficiently.
- To reduce manual errors in student management.
- To provide quick searching of student details.
- To display all student records easily.
- To remove unnecessary student records.
- To learn file handling concepts using C++.

3. System Features:

Main Features

1. Add New Student Record
2. Display All Student Records
3. Search Student by ID
4. Delete Student Record
5. Menu Driven User Interface
6. File Handling for Data Storage
7. Easy Console-Based Operation

4. Technologies and Concepts Used:

Technologies Used

- Programming Language: C++
- Platform: Console Application
- Compiler: Dev C++

Concepts Used

- Structures (struct)
- Functions
- File Handling
- Loops (while, do-while)
- Conditional Statements (if, switch)
- Character Arrays
- Input and Output Streams

5. System Design:

Structure Design

Structure: Student

DATA MEMBERS	DESCRIPTION
-----	-----
ID	Stores Student ID
NAME	Stores Student Name
MARKS	Stores Student Marks

Functions

FUNCTION	PURPOSE
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addStudent()	Adds student details

| displayStudents() | Displays all student records |

| searchStudent() | Searches student by ID |

| deleteStudent() | Deletes student record |

6. Workflow:

Step-by-Step Working Process

1. Program starts.
2. Main menu is displayed.
3. User selects an option.
4. Based on choice:
 - Add student record
 - Display student records
 - Search student
 - Delete student
5. System performs selected operation.
6. Menu repeats until user chooses Exit.
7. Program terminates.

7. Sample Output:

Main menu:

===== STUDENT MANAGEMENT SYSTEM =====

1. Add Student Record
2. Display Student Record
3. Search Student Record

4. Delete Student Record

5. Exit

Enter choice:1

Enter Name: Ravi

Enter Student ID: 101

Enter Marks: 89

Student Record Added Successfully!

Enter choice:2

Stuent Name : Ravi

Student ID : 101

Student Marks : 89

Enter choice:3

Enter Student ID to Search: 101

Student Found Successfully!

Enter choice:4

Enter Student ID to Delete: 101

Student Record Deleted Successfully!

Enter choice:5

THANK YOU

Conclusion:

☐ This project is a beginner-friendly application that demonstrates the use of C++ programming concepts, file handling, and menu-driven operations for managing student records efficiently.